Lesson plan · for the teacher

Checks that catch problems

Data engineering · Level 4 · Lesson 2

Time
40 min

Materials
A made-up mark or rain sheet with planted mistakes; Board and chalk; Optional: a phone or computer for the SQL steps

Learning goals

- Write checks: range, gaps, repeats, allowed values.
- Write a check as a query that returns the bad rows.
- Decide what a failure does, and fix the source.

Activity: Planted mistakes

1. Warm-up (5 min): Ask pupils to list mistakes a mark sheet could contain: a mark of 25, a blank, a pupil twice, a class that doesn't exist.
2. Teach (8 min): Turn each mistake into a check. Show one check as a SQL query that returns the bad rows.
3. Write (10 min): Groups write four checks for their class's marks, each as one clear rule.
4. Test (12 min): Give each group a made-up sheet with five planted mistakes, one of them believable. Groups run their checks and note what they found and missed.
5. Wrap-up (5 min): Reveal the believable mistake no check caught, and discuss how a person would notice it. Then give the exit ticket.

Plant one mistake of each kind, plus one believable one (such as 12 typed as 15) that no rule can catch. Use made-up names.

Discussion

- Which check would have stopped the 999?
- Why is "= NULL" a check that can never fail?
- What should we say when every check passes?

Exit ticket

- Write one check as a rule and as a SQL query, and what should happen when it fails.

Try it at school tomorrow
Ask your pupils to write checks for their own class data: a mark must be from 0 to 20, every pupil appears once. Then test the checks on a made-up sheet with planted mistakes.

Practice questions · for you

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1 **Every check passes on today's batch. What can you say about the data?**

◯ No known problem was found ◯ It's correct ◯ The checks are broken

2 **A range check as a query: find every reading outside 0 to 300 mm. Show the station, the day and the amount.**

```
SELECT station, day, mm FROM rain
WHERE mm > 300;
```

Write your program:

3 **Which of these faults would the range check (0 to 300 mm) let through? Choose all that apply.**

☐ A reading sent twice ☐ 12 mm typed when it really rained 21 ☐ 999, the error code

☐ A minus sign: -3 ☐ An empty reading

4 **A uniqueness check: find every station and day that has more than one reading. Show the station, the day and how many readings it has.**

```
SELECT station, day, COUNT(*) FROM rain
GROUP BY station, day;
```

Write your program:

5 **This check should find the empty readings, but it never returns a row, even when Lakeside's reading is missing. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  SELECT station, day
2  FROM rain
3  WHERE mm = NULL;
```

✎ **Pick one dataset your school sends out. Which four checks would you run on it, and what should happen when each one fails?**

Check your answers

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1 **No known problem was found**

Checks find only the problems you thought of.

2

```
SELECT station, day, mm FROM rain
WHERE mm < 0 OR mm > 300;
```

A reading is out of range if it's below 0 or above 300, so the two conditions are joined with OR: Riverside's 999 and Market's -3.

3 **12 mm typed when it really rained 21; A reading sent twice; An empty reading**

A range check catches only amounts outside the range, here 999 and -3.

4

```
SELECT station, day, COUNT(*) FROM rain
GROUP BY station, day
HAVING COUNT(*) > 1;
```

GROUP BY station, day makes one group per reading slot; HAVING COUNT(*) > 1 keeps the slots with repeats: Hilltop on Wednesday, twice.

5 **Line 3:** `WHERE mm IS NULL;`

Nothing equals NULL, not even NULL, so = NULL is never true and the check always "passes".